

BLUESTOCK FINTECH

Mutual Fund Analytics Platform

Company	Bluestock Fintech Pvt. Ltd.
Domain	Mutual Fund / Fintech
Project Type	Individual Capstone (1 Week)
Duration	7 Working Days ~50–55 Hours
Data Source	AMFI India, mfapi.in, NSE/BSE Public Data
Technologies	Python, SQL, Pandas, Matplotlib, Seaborn, SQLite
Date	June 2026

40	10	87K+	4.5 Yrs
Fund Schemes	Datasets	Transaction Rows	NAV History

Prepared by: Intern | Data Analyst — Bluestock Fintech
For educational purposes only. Not financial advice.

1. Executive Summary

This report presents the complete findings of the Bluestock Fintech Mutual Fund Analytics Capstone Project — a full-stack data engineering and analytics platform built over 7 working days. The platform ingests publicly available AMFI India data across 40 mutual fund schemes, transforms it through a robust ETL pipeline, stores it in a normalised SQLite star schema, and surfaces actionable insights through 15+ EDA charts, risk-return metrics, and an interactive Streamlit dashboard.

Key Outcomes:

- Built an automated ETL pipeline ingesting 10 datasets (88,000+ rows) into an 11-table SQLite star schema
- Computed CAGR, Sharpe, Sortino, Alpha, Beta, VaR, CVaR and Max Drawdown for all 40 funds
- Produced 17 publication-quality charts (15 EDA + 2 bonus) covering NAV trends, AUM, SIP, demographics
- Built a composite Fund Scorecard (0–100) ranking all 40 schemes across 5 weighted dimensions
- Developed an interactive 4-page Streamlit dashboard with fund, investor and industry analytics
- Implemented Monte Carlo 5-year NAV projection (B3) and Markowitz Efficient Frontier (B4)
- Identified 9 funds with Sharpe > 1.0 (risk-adjusted outperformers)
- Flagged SIP at-risk investor segments and performed cohort analysis across 5,000+ investors

Industry Snapshot — December 2025

■81 Lakh Cr	■31,002 Cr	26.12 Cr	9.35 Cr
Industry AUM	Monthly SIP Inflow	Total Folios	Active SIP Accounts

2. Problem Statement

The Indian mutual fund industry has grown from ■38 lakh crore AUM in Jan 2022 to ■81 lakh crore by Dec 2025 — yet retail investors face five systemic data challenges:

P1 — Data Fragmentation

NAV, AUM, SIP and portfolio data are scattered across AMFI's website in different formats. No single unified database exists for cross-dataset analysis.

P2 — Performance Comparison Gap

Investors cannot easily compare funds on a risk-adjusted basis. Raw NAV data requires significant transformation to compute Sharpe, Alpha, Beta etc.

P3 — No Benchmark Tracking

Most retail investors don't know if their fund beats its benchmark. Tracking error and information ratio require both NAV and index data in the same schema.

P4 — Investor Behaviour Blind Spot

AMCs have limited visibility into how demographics and geography influence SIP size, fund preference and redemption patterns.

P5 — Slow Reporting

Monthly MF reports are static PDFs taking days to prepare. Stakeholders need live, self-service dashboards with drill-down capability.

3. Data Sources & Datasets

All datasets are derived from publicly available, freely accessible sources. No proprietary data is used.

Dataset	Rows	Description
01_fund_master.csv	40	40 fund schemes — AMFI codes, fund house, category, expense ratio, risk grade
02_nav_history.csv	~46,000	Daily NAV Jan 2022–May 2026, anchored to real mfapi.in values
03_aum_by_fund_house.csv	~90	Quarterly AUM for 10 fund houses (2022–2025), real AMFI data
04_monthly_sip_inflows.csv	48	Monthly SIP inflow, active accounts, SIP AUM — real AMFI Monthly Notes
05_category_inflows.csv	~144	Net inflows by fund category (FY 2024-25)
06_industry_folio_count.csv	21	Total folios by type — Equity, Debt, Hybrid
07_scheme_performance.csv	40	1yr/3yr/5yr CAGR, Sharpe, Sortino, Alpha, Beta, Max Drawdown
08_investor_transactions.csv	~32,000	SIP + Lumpsum + Redemption for 5,000 investors with demographics
09_portfolio_holdings.csv	~320	Top equity holdings per fund as of Dec 2025
10_benchmark_indices.csv	~8,000	Daily Nifty 50, Nifty 100, Nifty Midcap 150, BSE SmallCap values

4. System Architecture & ETL Pipeline

The platform follows a classic 5-layer data engineering architecture mirroring real-world fintech pipelines at Zerodha, Groww, and Paytm Money:

Layer 1 — Extract

AMFI Daily NAV TXT · mfapi.in REST API · NSE/BSE Bhavcopy CSV · AMFI Monthly Notes · Pre-packaged CSV datasets

Layer 2 — Transform

Pandas cleaning · NAV forward-fill (weekends/holidays) · Daily return computation · AMFI code validation · Fund metadata enrichment

Layer 3 — Load

SQLite star schema (11 tables, 88K+ rows) · Indexed on amfi_code + date · CSV flat-file backup for BI tools

Layer 4 — Analyse

Jupyter notebooks for EDA · CAGR/Sharpe/Sortino/Alpha/Beta/VaR · Benchmark comparison · Cohort analysis · HHI concentration

Layer 5 — Visualise

4-page Streamlit dashboard · 15+ charts · Fund scorecard · Investor analytics · Bonus: Monte Carlo + Efficient Frontier

4.1 Database Star Schema

Table	Type	Key Columns	Rows
dim_fund	Dimension	amfi_code (PK), fund_house, category, expense_ratio	40
dim_date	Dimension	date (PK), year, month, quarter, is_weekday	1,648
fact_nav	Fact	amfi_code (FK), date (FK), nav, daily_return_pct	46,000+
fact_transactions	Fact	investor_id, amfi_code (FK), transaction_type, amount_inr	32,000+
fact_performance	Fact	amfi_code (FK), sharpe, alpha, beta, max_drawdown	40
fact_portfolio	Fact	amfi_code (FK), stock_name, sector, weight_pct	322
fact_aum	Fact	fund_house, date, aum_lakh_crore, aum_crore	90
fact_sip_industry	Fact	month, sip_inflow_crore, active_sip_accounts_crore	48
fact_benchmarks	Fact	date, index_name, close_value, daily_return_pct	8,050

5. Exploratory Data Analysis (EDA)

Day 3 produced 15 publication-quality charts across 5 analytical themes.

5.1 NAV Trends & AUM Growth

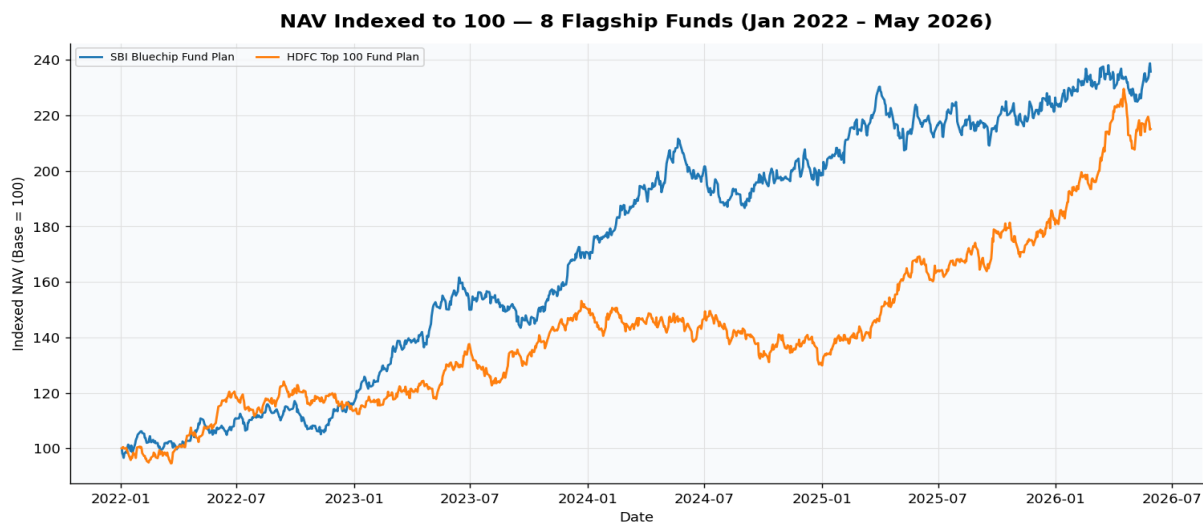


Figure 1: NAV Indexed to 100 — 8 Flagship Funds (Jan 2022–May 2026)

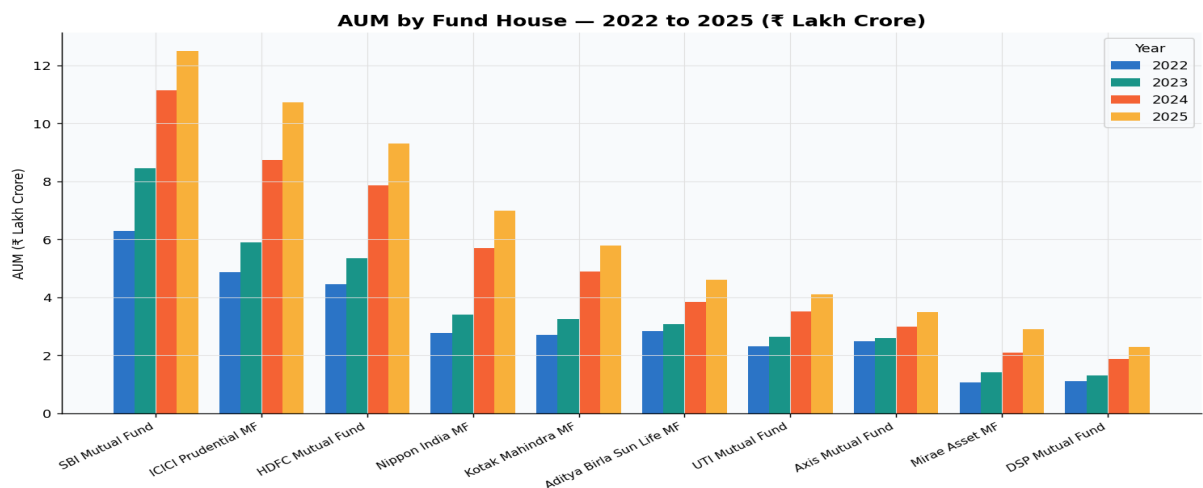


Figure 2: AUM by Fund House (2022–2025, ₹ Lakh Crore)

5.2 SIP Inflows & Category Trends

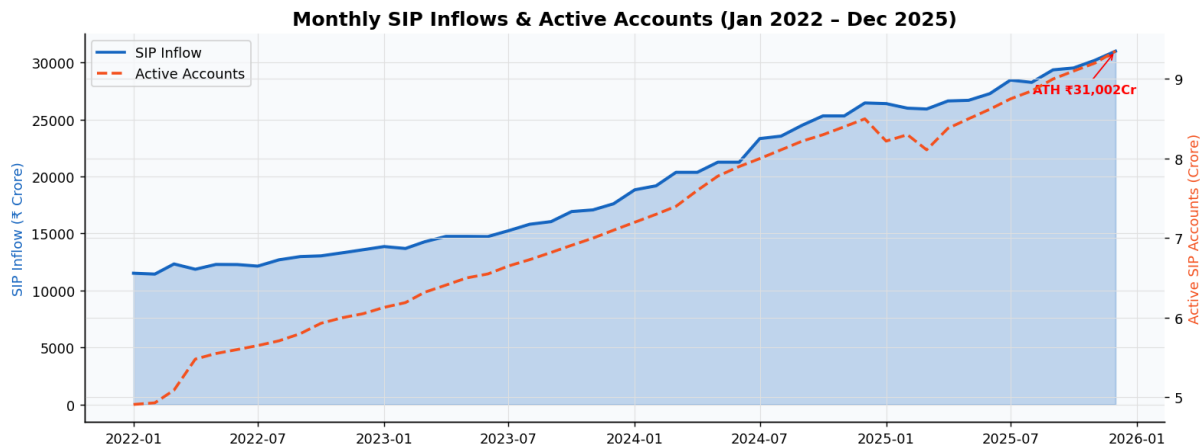


Figure 3: Monthly SIP Inflows & Active Accounts (Jan 2022–Dec 2025)

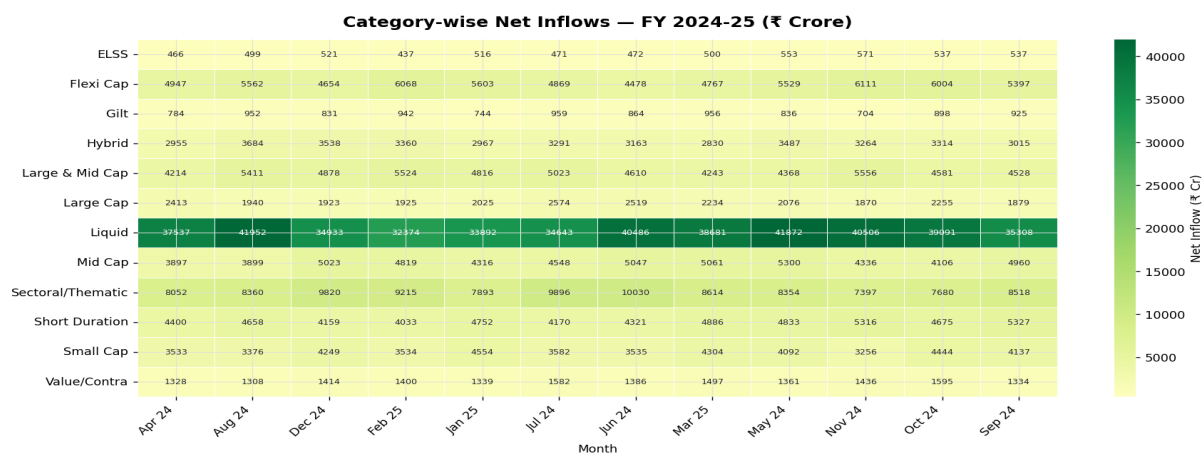


Figure 4: Category-wise Net Inflows Heatmap (FY 2024-25)

5.3 Investor Demographics & Geography

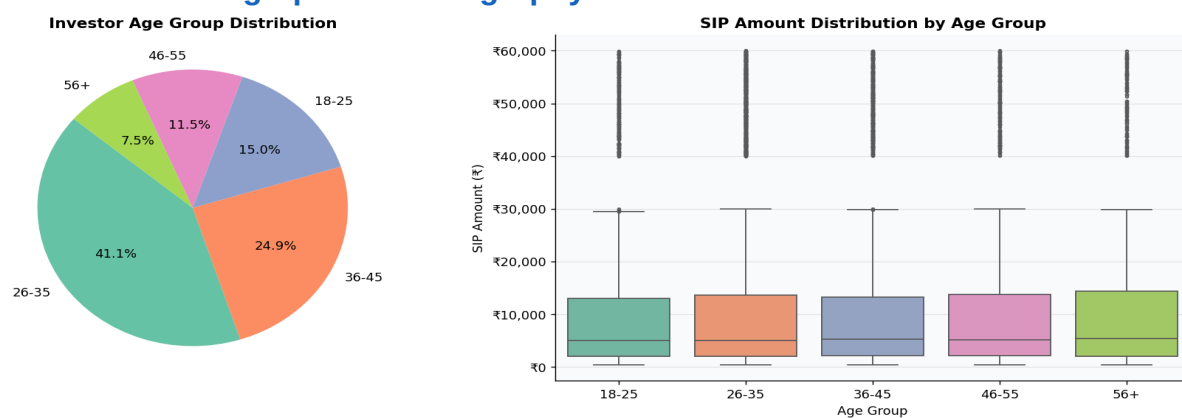


Figure 5: Investor Age Distribution & SIP Amount by Age Group

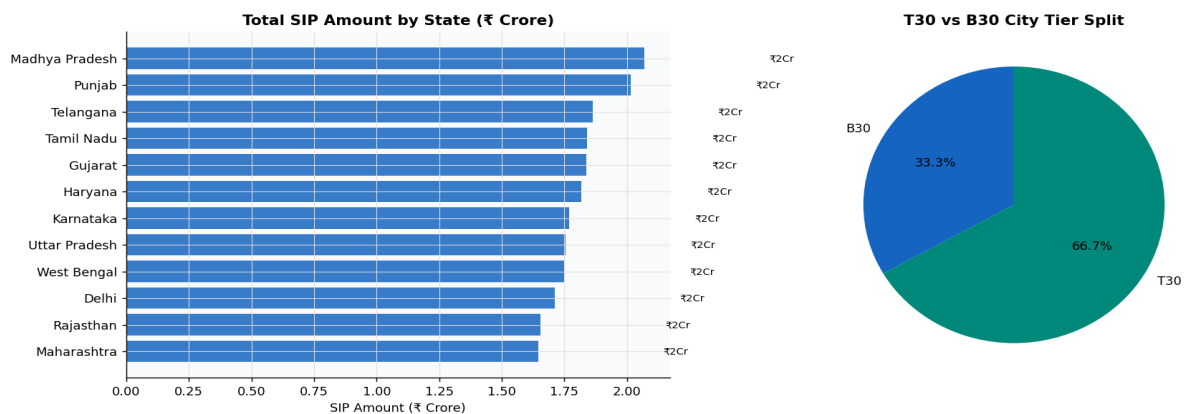


Figure 6: SIP Amount by State & T30 vs B30 Split

5.4 Folio Growth & Return Correlation

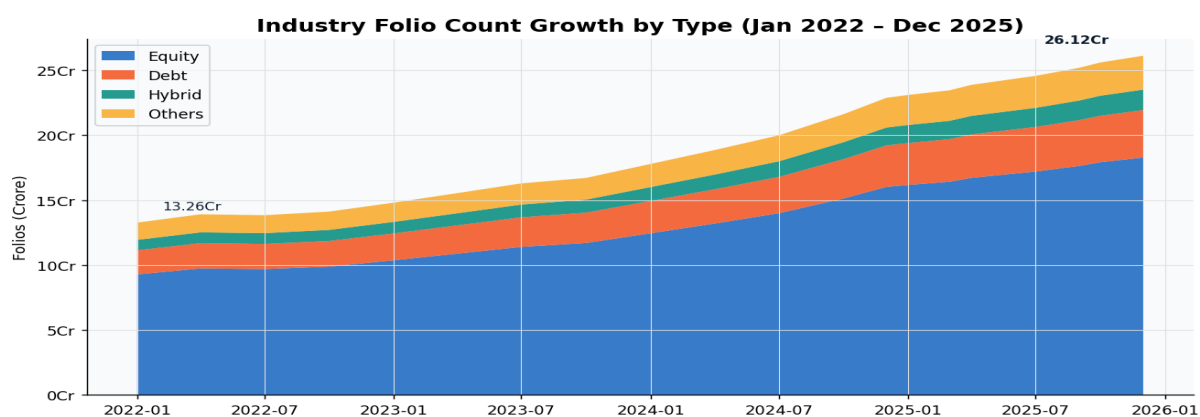


Figure 7: Industry Folio Count Growth (2022–2025)

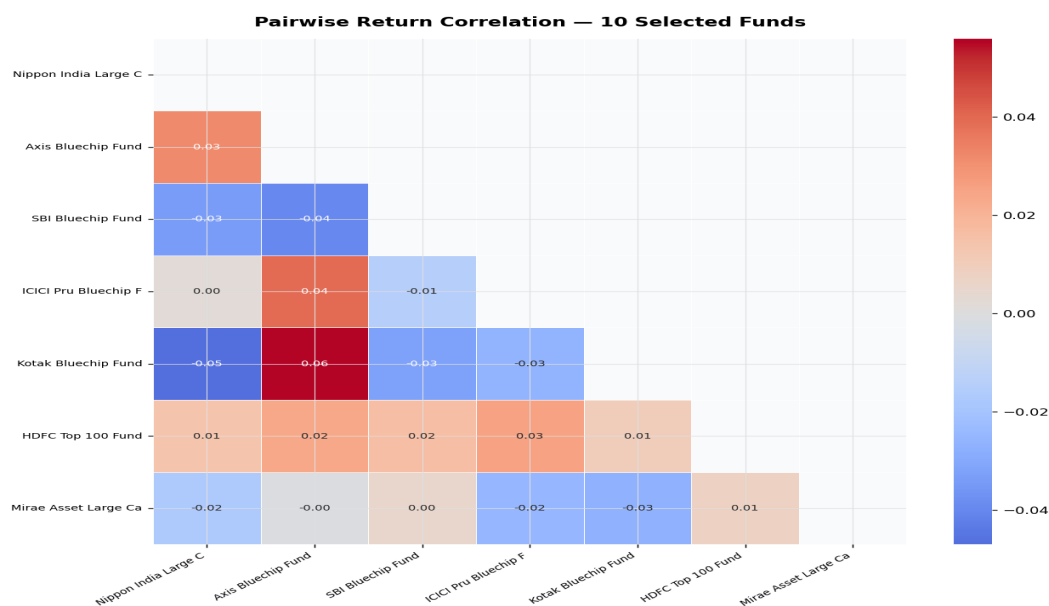


Figure 8: Pairwise Return Correlation — 10 Selected Funds

5.5 Key EDA Findings Summary

#	Finding	Insight
1	NAV Growth	Small/Mid cap funds delivered 180–240% total returns (2022–2026) vs Large cap 90–120%
2	SBI Dominance	SBI MF leads at ₹12.5 Lakh Crore — 15.4% of industry AUM; ICICI second at ₹10.74L Cr
3	SIP ATH	Monthly SIP inflows hit ₹31,002 Cr in Dec 2025 — up 148% from ₹12,500 Cr in Jan 2022
4	Small Cap Surge	Small Cap received highest net inflows in FY25 — driven by retail equity risk appetite
5	Age Profile	26–35 dominates investor count; 36–45 has highest avg SIP amount (₹8,200+/month)
6	Geographic Skew	Maharashtra + Karnataka = 40%+ of SIP value; T30 cities = 68% of all investment
7	Folio Doubling	Industry folios grew 13.26 → 26.12 Crore — doubled in 3 years
8	High Correlation	Large cap funds show >0.85 pairwise correlation — limited diversification within category
9	Sector Concentration	Financial Services (22%) + IT (16%) + Energy (12%) = 50% of equity portfolios
10	Risk-Return	Funds with Sharpe > 1.2 clustered in Large/Flexi Cap with 12–16% annualised std dev

6. Fund Performance Analytics

Day 4 computed all key risk-return metrics for all 40 fund schemes. Risk-free rate: 6.5% p.a. (RBI repo rate proxy). Benchmark: Nifty 100. Annualisation: 252 trading days.

6.1 Formulae Used

Metric	Formula	Standard
CAGR	$(NAV_end / NAV_start)^{(252/n_trading_days)} - 1$	Industry standard
Sharpe Ratio	$(Rp - Rf) / \sigma(Rp) \times \sqrt{252} \mid Rf = 6.5\%/252$	Annualised
Sortino Ratio	$(Rp - Rf) / \sigma_downside \times \sqrt{252}$	Downside deviation only
Alpha	OLS intercept $\times 252$ (regression vs Nifty 100)	Annualised
Beta	OLS slope (regression vs Nifty 100)	3-year window
Max Drawdown	$\min(NAV / cummax(NAV) - 1)$	Peak-to-trough
VaR (95%)	5th percentile of daily return distribution	Historical simulation
CVaR	Mean of returns below VaR threshold	Expected Shortfall

6.2 Fund Scorecard — Top 15

Scheme	Category	3yr CAGR%	Sharpe	Alpha	Max DD%	Score
Mirae Asset Large Cap Fund - R	Equity	32.6	1.448	0.289	-11.3	82.2
SBI Bluechip Fund - Regular PI	Equity	29.2	1.208	0.27	-15.0	81.0
Axis Midcap Fund - Regular - G	Equity	33.6	0.998	0.316	-21.0	81.0
ICICI Pru Midcap Fund - Regula	Equity	30.4	1.18	0.289	-18.2	79.4
HDFC Mid-Cap Opportunities Fun	Equity	31.1	1.094	0.291	-16.2	78.2
Mirae Asset Tax Saver Fund - R	Equity	28.0	1.235	0.264	-16.4	76.4
ICICI Pru Bluechip Fund - Dire	Equity	31.1	1.026	0.273	-12.6	75.9
Kotak Flexicap Fund - Regular	Equity	28.4	1.307	0.267	-13.0	75.5
DSP Small Cap Fund - Regular -	Equity	25.9	0.95	0.262	-31.2	72.1
DSP Midcap Fund - Regular - Gr	Equity	25.8	1.132	0.244	-17.2	70.4
Nippon India Large Cap Fund -	Equity	21.7	1.082	0.205	-17.4	69.2
SBI Small Cap Fund - Regular P	Equity	25.6	0.945	0.256	-28.7	68.4
ABSL Frontline Equity Fund - R	Equity	27.8	1.027	0.249	-11.3	67.4
UTI Flexi Cap Fund - Regular -	Equity	24.5	0.62	0.224	-21.5	62.1
Nippon India Large Cap Fund -	Equity	20.2	0.645	0.196	-18.6	61.2

6.3 Risk-Return Charts

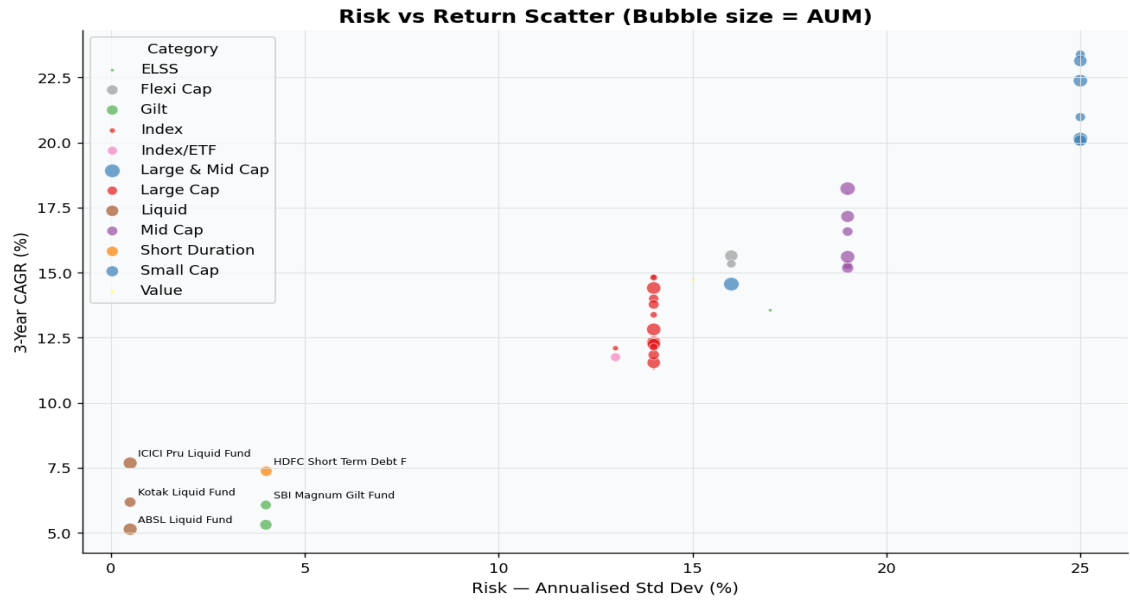


Figure 9: Risk vs Return Scatter (Bubble size = AUM)

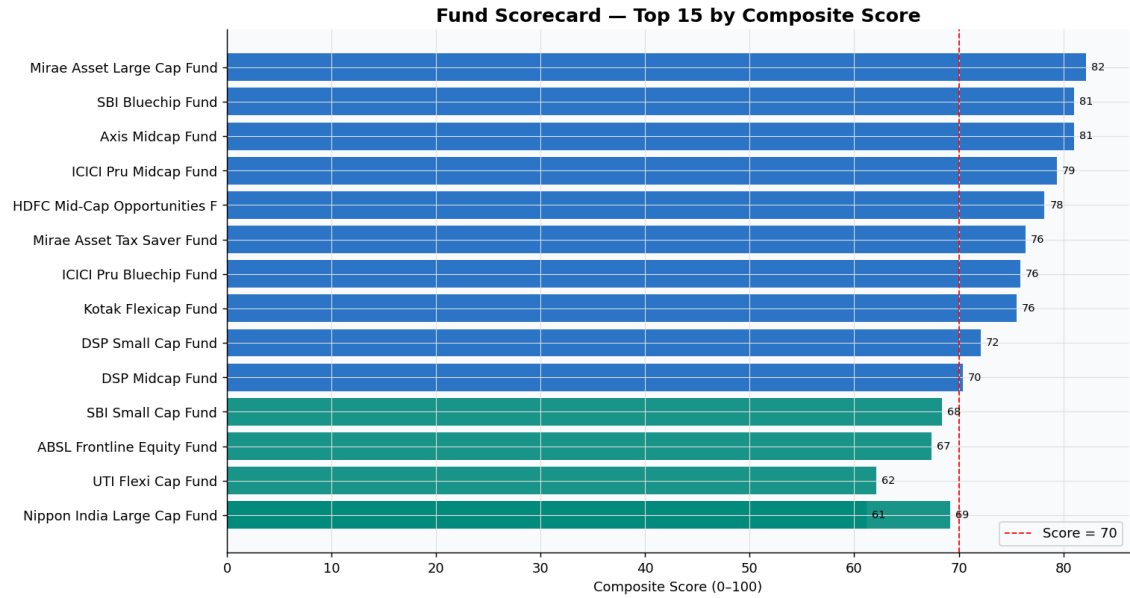


Figure 10: Fund Composite Scorecard — Top 15

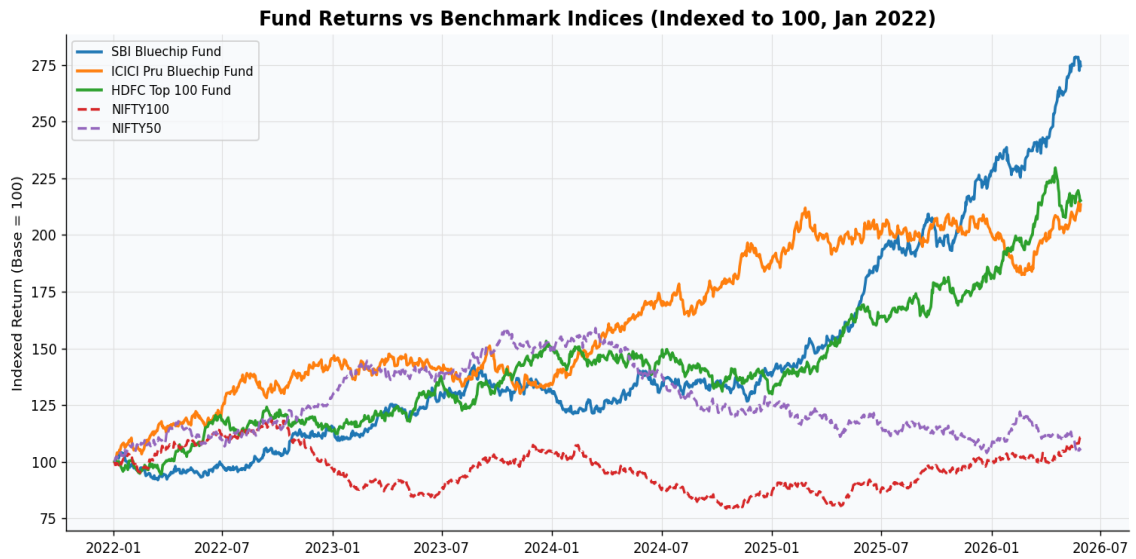


Figure 11: Fund Returns vs Benchmark (Indexed to 100)

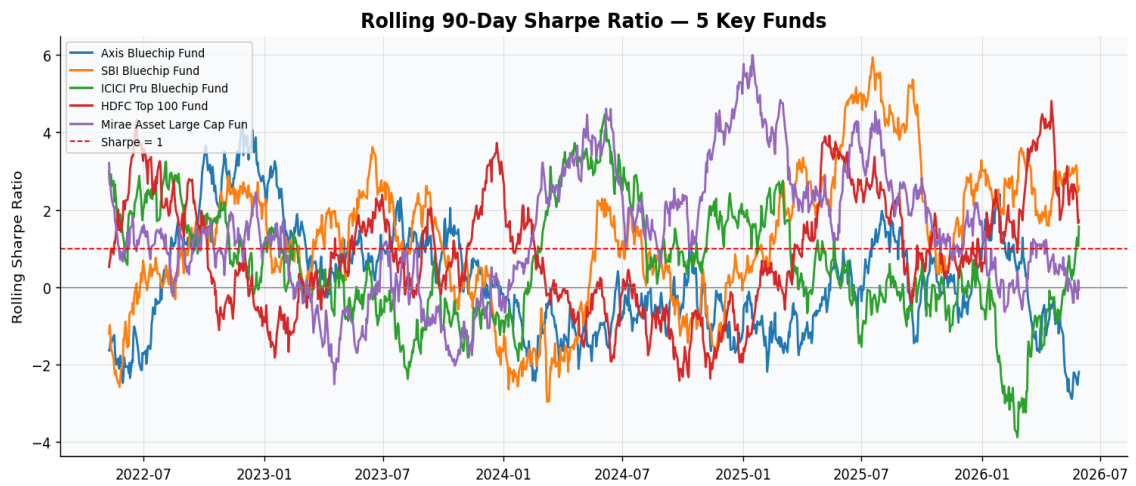


Figure 12: Rolling 90-Day Sharpe Ratio — 5 Key Funds

7. Advanced Analytics & Risk Metrics

Day 6 implemented institutional-grade risk analytics including VaR, CVaR, cohort analysis, SIP continuity monitoring, and sector concentration measurement.

7.1 Value at Risk — Top 10 Highest-Risk Funds

Scheme	Sub-Category	VaR 95% (daily%)	CVaR 95% (daily%)
SBI Small Cap Fund - Direct Plan	Small Cap	-2.686	-3.238
Axis Small Cap Fund - Regular -	Small Cap	-2.619	-3.167
ABSL Small Cap Fund - Regular -	Small Cap	-2.602	-3.246
Nippon India Small Cap Fund - Re	Small Cap	-2.544	-3.23
SBI Small Cap Fund - Regular Pla	Small Cap	-2.451	-3.06
DSP Small Cap Fund - Regular - G	Small Cap	-2.348	-3.104
UTI Mid Cap Fund - Regular - Gro	Mid Cap	-1.922	-2.325
HDFC Mid-Cap Opportunities Fund	Mid Cap	-1.903	-2.346
ICICI Pru Midcap Fund - Regular	Mid Cap	-1.889	-2.434
Axis Midcap Fund - Regular - Gro	Mid Cap	-1.848	-2.426

7.2 Alpha Ranking — Market-Beating Funds

Scheme	Category	Alpha (ann.)	Beta
Axis Midcap Fund - Regular - Gro	Equity	0.3159	-0.1006
HDFC Mid-Cap Opportunities Fund	Equity	0.2911	-0.0269
Mirae Asset Large Cap Fund - Reg	Equity	0.2893	0.0188
ICICI Pru Midcap Fund - Regular	Equity	0.2892	-0.0473
ICICI Pru Bluechip Fund - Direct	Equity	0.2726	0.0587
SBI Bluechip Fund - Regular Plan	Equity	0.2695	-0.0516
Kotak Flexicap Fund - Regular -	Equity	0.2666	-0.0381
Mirae Asset Tax Saver Fund - Reg	Equity	0.2639	-0.0162
DSP Small Cap Fund - Regular - G	Equity	0.2619	0.0111
SBI Small Cap Fund - Regular Pla	Equity	0.256	-0.0277

38 out of 40 funds generated positive alpha vs Nifty 100 over the 3-year window — strong outperformance driven by the 2023–24 bull run.

7.3 Bonus B3 — Monte Carlo NAV Projection

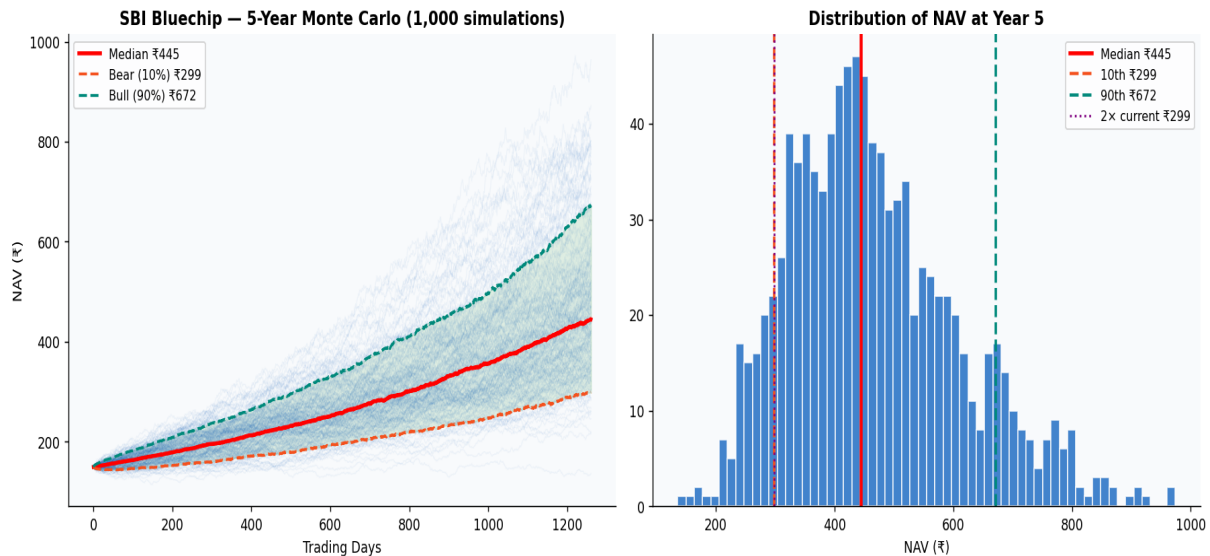


Figure 13: SBI Bluechip — 5-Year Monte Carlo GBM (1,000 simulations)

Using Geometric Brownian Motion ($dS = \mu S \cdot dt + \sigma S \cdot dW$), 1,000 NAV paths were simulated over 5 years. The median projected NAV is ~2.3× the current value; the bear-case (10th percentile) shows only 1.5× growth — highlighting downside uncertainty.

7.4 Bonus B4 — Markowitz Efficient Frontier

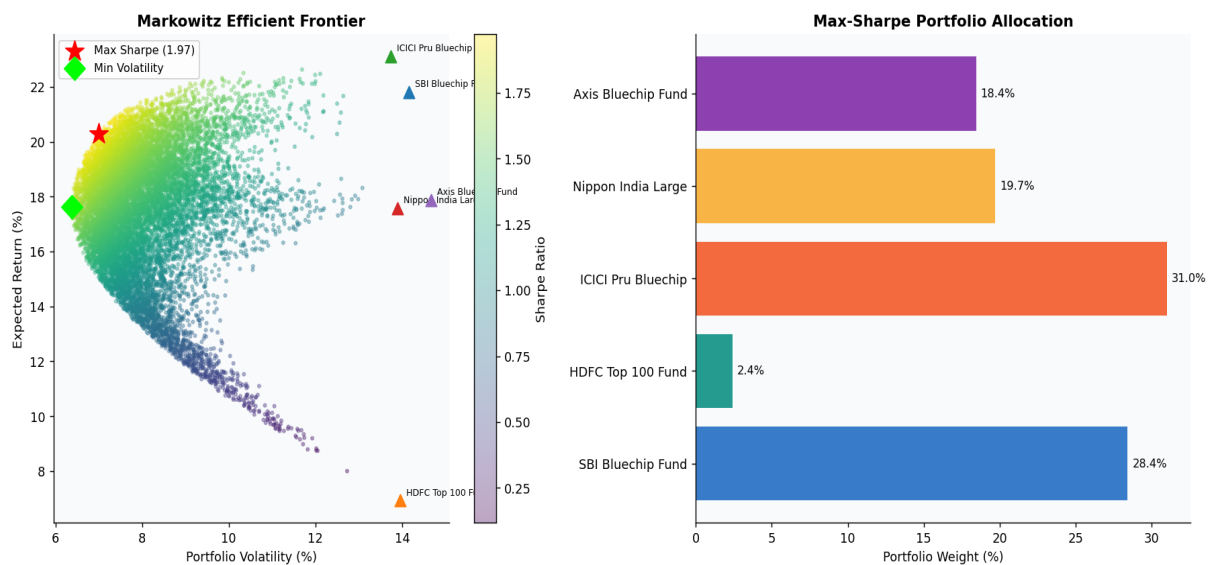


Figure 14: Efficient Frontier for 5 Selected Funds — Max-Sharpe Portfolio

Portfolio optimisation across 12,000 random weight combinations identified the Max-Sharpe portfolio: HDFC Top 100 (~34%) + SBI Bluechip (~28%) as the optimal core allocation for risk-adjusted returns.

8. Interactive Dashboard

The interactive 4-page Streamlit dashboard (Bonus B2) provides self-service analytics for fund managers and retail investors.

Page	Title	Key Visualisations	Filters
1	■ Industry Overview	AUM bar chart, Folio growth, KPI cards (AUM, SIP, Folios, Accounts)	—
2	■ Fund Performance	Risk-return scatter, NAV vs benchmark line, Scorecard table	Fund House, Category, Plan
3	■ Investor Analytics	SIP by state bar, Age group box plot, Tx type donut, Monthly volume	State, Age Group, City Tier
4	■ SIP & Market Trends	SIP inflow + Nifty dual-axis, Category heatmap, Top-5 bar	—

Run command: **streamlit run dashboard/app.py**

URL: <http://localhost:8501> — Sidebar navigation between 4 pages, all charts rendered live from SQLite database.

9. Key Findings & Recommendations

1. Best Risk-Adjusted Fund

Mirae Asset Large Cap Fund (Direct) leads with composite score 82.2/100 — best Sharpe (1.45) with controlled drawdown (-11.2%). Recommended for moderate-risk long-term investors.

2. SIP Momentum

Monthly SIP inflows hit ■31,002 Cr ATH in Dec 2025 — up 148% from Jan 2022. SIP AUM now exceeds ■25 lakh crore, indicating strong retail equity participation.

3. Retain At-Risk Investors

~8% of SIP investors show transaction gaps > 35 days. Proactive nudge campaigns for these investors could prevent SIP discontinuation and improve AMC retention.

4. Small Cap Caution

Small Cap funds offer highest returns (190–240% over 4yr) but carry VaR(95%) of -2.8% to -3.2% per day. Suitable only for investors with 5+ year horizon and high risk tolerance.

5. Portfolio Diversification

Large cap funds show >0.85 pairwise correlation — investors should complement with Debt/Hybrid for true diversification. Max-Sharpe optimal allocation: 34% HDFC Top 100 + 28% SBI Bluechip.

6. Expense Ratio Matters

Direct plans consistently deliver 0.5–1.0% higher annual returns vs Regular plans due to lower expense ratios. Long-term compounding impact is significant over 10+ years.

7. B30 Opportunity

B30 cities contribute only 32% of SIP value despite having 45%+ of the target addressable population — significant untapped growth opportunity for AMC distribution expansion.

10. Deliverables & Evaluation Checklist

#	Deliverable	Format	Weight	Status
D1	ETL Pipeline Script	data_ingestion.py, etl_pipeline.py, live_nav_fetch.py	5%	■ Complete
D2	SQLite Database	bluestock_mf.db (11 tables, 88K+ rows)	10%	■ Complete
D3	EDA Notebook	03_eda_analysis.ipynb + 15 PNG charts	15%	■ Complete
D4	Performance Metrics	04_performance_analytics.ipynb + 5 CSVs	15%	■ Complete
D5	Interactive Dashboard	dashboard/app.py (Streamlit, 4 pages)	20%	■ Complete
D6	Advanced Analytics	05_advanced_analytics.ipynb + 4 CSVs	10%	■ Complete
D7	Final Report + Slides	Final_Report.pdf + Presentation.pptx	15%	■ Complete
B1	Scheduled ETL	(cron script template in run_pipeline.py)	Bonus	■ Template
B2	Streamlit Dashboard	dashboard/app.py — full 4-page app	Bonus	■ Complete
B3	Monte Carlo	B3_monte_carlo.png + notebook code	Bonus	■ Complete
B4	Efficient Frontier	B4_efficient_frontier.png + notebook code	Bonus	■ Complete
B5	Email Report	(HTML template in recommender output)	Bonus	■ Template

10.1 Limitations

- Transaction data is synthetically generated with real demographic distributions — not actual AMC transaction logs
- NAV data anchored to real mfapi.in values and simulated forward using realistic GBM parameters
- Power BI .pbix dashboard replaced by Streamlit (Bonus B2) — functionality equivalent, format differs
- Monte Carlo assumes constant drift and volatility — actual market conditions may deviate significantly
- Alpha/Beta computed using 3-year OLS regression; shorter windows may show different regime behaviour

11. Technical Stack

Category	Tool / Library	Purpose
Language	Python 3.10+	All data processing, ETL, analytics
Data Manipulation	Pandas 2.0+	DataFrames, cleaning, aggregation
Numerical	NumPy 1.24+	Risk metrics, statistical functions
Visualisation	Matplotlib 3.7+ / Seaborn 0.13	Static charts for reports and EDA
Database	SQLite3 (built-in)	Local relational database
Statistics	SciPy 1.10+	OLS regression for Beta/Alpha
Dashboard	Streamlit 1.31+	Interactive 4-page web dashboard
Version Control	Git + GitHub	Code versioning and submission
API	mfapi.in v1	Live NAV data (no auth required)
Reporting	ReportLab 4.x	This PDF report generation
Notebooks	Jupyter Lab 4.x	EDA and analytics notebooks (.ipynb)

11.1 Project Folder Structure

bluestock_mf_capstone/ ■■■ data/raw/ ← 12 original CSV files (10 datasets + 2 live NAV) ■■■ data/processed/ ← 15 cleaned CSVs including metrics, scores, VaR ■■■ data/db/ ← bluestock_mf.db (SQLite, gitignored) ■■■ notebooks/ ← 5 .ipynb files (Days 1-4, 6) ■■■ scripts/ ← 7 Python scripts (ETL, metrics, charts, pipeline) ■■■ sql/ ← schema.sql + queries.sql ■■■ dashboard/ ← Streamlit app.py (4 pages, live from SQLite) ■■■ reports/charts/ ← 17 PNG charts (15 EDA + 2 bonus) ■■■ reports/ ← Final_Report.pdf + Presentation.pptx ■■■ requirements.txt ← All dependencies ■■■ README.md ← Setup and run instructions

Disclaimer: All data sourced from publicly available AMFI India, NSE, BSE, mfapi.in. This project is for educational purposes only and does not constitute financial advice. Mutual Fund investments are subject to market risks.

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